

Learning Objective 3.9: I can calculate null-steering weights and implement pattern nulls on the ADALM-PHASER.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Why a null beats gain.** The PHASER views a target on boresight while an interferer sits at $+22.5^\circ$. At the aperture the interferer is 10 dB stronger than the target. The array is uniformly weighted and steered broadside, so its pattern level at $+22.5^\circ$ is the first sidelobe level, -13 dB relative to the peak.

- (a) Compute the signal-to-interference ratio at the beam-former output.

SIR =

- (b) Null steering replaces the sidelobe at $+22.5^\circ$ with a notch measured at -21.6 dB relative to the new main-lobe peak. Recompute the SIR.

SIR =

- (c) The same 8.6 dB of margin could be bought by raising the transmit power. State what that costs on a radar, in one or two sentences, and say which fix you would specify.

- (d) The interferer is at $+22.5^\circ$ and the target is on boresight. Explain why the array does not reject the interferer simply because it is off axis.

2. **Steering vectors.** The PHASER array has $N = 8$ elements at $d = 14$ mm, receiving the HB100 tone at 10.525 GHz so that $d/\lambda = 0.491$ and $kd = 3.085$ rad. The steering vector for arrival angle θ has entries $v_n(\theta) = e^{jnkd\sin(\theta)}$, and the beamformer response is $y(\theta) = \sum_n w_n v_n(\theta)$.

(a) In one or two sentences, state what $\vec{v}(\theta)$ describes and what it does *not* depend on.

- (b) Compute the element-to-element phase for a commanded beam at $\theta_0 = 30^\circ$.

$\Delta\phi =$

- (c) Write the weight w_3 (the fourth element, $n = 3$) that steers the beam to 30° , as a magnitude and an angle.

$w_3 =$

- (d) With those weights loaded, evaluate $y(30^\circ)$ and express the result in dB relative to one element.

$y =$

3. **Weight subtraction by hand.** A four-element array with $d = \lambda/2$ (so $kd = 180^\circ$) is steered broadside, giving $\vec{w}_d = [1, 1, 1, 1]$. An interferer appears at $\theta_1 = 45^\circ$. Use the weight-subtraction rule $\vec{w} = \vec{w}_d - r_n \vec{w}_n$ with $r_n = \vec{w}_n^H \vec{w}_d / \vec{w}_n^H \vec{w}_n$.

- (a) Compute the element-to-element phase $\psi_1 = kdsin(\theta_1)$ at the interferer angle.

$$\psi_1 = \boxed{}$$

- (b) Evaluate r_n . For a broadside beam it reduces to the normalized uniform array factor at θ_1 , $r_n = \frac{e^{j(N-1)\psi_1/2} \sin(N\psi_1/2)}{(N \sin(\psi_1/2))}$.

$$r_n = \boxed{}$$

- (c) Compute the four weights $w_n = 1 - r_n e^{-jn\psi_1}$ and convert them to PHASER-style settings: element gains as percentages of the largest, and phases in degrees.

- (d) Find the loss in the look direction after the weights are rescaled so the largest element sits at 100% gain.

$$\text{loss} = \boxed{}$$

4. **What the null costs.** Return to the eight-element PHASER array ($kd = 3.085$ rad, broadside beam). Before the weights are rescaled, the response in the look direction falls from N to $N(1 - |r_n|^2)$.

(a) An interferer sits at $\theta_1 = 10^\circ$, where the uniform pattern level is -8.0 dB. Compute the loss in the look direction before rescaling.

loss =

(b) A second interferer sits at $\theta_1 = 30^\circ$, where the uniform pattern level is -33.9 dB. Compute the loss and comment on the result in one sentence.

loss =

(c) The PHASER's half-power beamwidth is 13.1° . A student asks the array to null an interferer at $\theta_1 = 3^\circ$, where $|r_n| = 0.93$. Explain what the resulting pattern looks like and what the student should do instead.

(d) Explain, in one or two sentences, why $|r_n|$ is worth computing before any weights are calculated.

5. **How deep can the null be?** Each ADAR1000 phase setting lands on a 2.8125° grid and each gain on a 1% grid, so each weight carries an independent error. The errors add in RMS at the null angle, giving a residual response $|y(\theta_1)|/N \approx \epsilon_{\text{rms}}/\sqrt{N}$ with $\epsilon_{\text{rms}} = \sqrt{\sigma_\phi^2 + \sigma_a^2}$ and $\sigma_\phi = \text{LSB}/\sqrt{12}$ in radians.

- (a) Compute the quantization-limited null depth for the eight-element PHASER with its 7-bit phase shifters.

depth =

- (b) Repeat for a 3-bit phase shifter, ignoring the gain error. Compare your answer with the $-6B$ rule of thumb from L26.

depth =

- (c) The PHASER carries 7-bit phase shifters, yet the lab sweep measures a notch of only -21.6 dB relative to the peak. Explain the discrepancy, given that the sweep's noise floor sits 23 dB below the uniform-taper peak and the null-steered main lobe sits 2 dB below that same reference.

- (d) State the maximum number of independent nulls an eight-element array can place and explain the counting in one sentence.

Documentation: _____